Roll No.....

Mid Semester Examination September 2025

M. Tech (VLSI Design and Systems) | Semester: IIIrd
VLSI Physical Design Automation | Paper Code: VDM-301

Time: 90 Minutes | Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Write a short note on full custom design and standard cell design	**CO1**
	OR	**CO2**
(b)	What are the main stages of the VLSI physical design cycle?	
Q2	(10 marks)	
(a)	What is the role of EDA tools in the VLSI design cycle? Discuss few EDA tool.	**CO1**
	OR	**CO2**
(b)	Discuss the transistor level design for AB+BC(E+F).	
Q3	(10 marks)	
(a)	Discuss the Fabrication Process in VLSI and its Impact on Physical Design.	**CO1**
	OR	**CO2**
(b)	What are the verifications method used in VLSI design.	
Q4	(10 marks)	
(a)	Discuss CMOS design Rule. Draw the stick diagram for CMOS NAND and NOR gate.	**CO1**
	OR	
(b)	What are the types of compaction techniques used in VLSI.	**CO2**
Q5	(10 marks)	
(a)	What are the algorithm used in placement process? Explain.	**CO1**
	OR	**CO2**
(b)	What are the algorithm used in partitioning process? Explain.	